TurboQuant v1.6.0

Polar Derotate + Tangent Residual Breakthrough

2026-04-14 ~ 04-15 | Qwen3-14B | TBQX3_1 · 3.625 bpw

Math +20% vs f16 | 4.74x compression | v1.6.0 release | AmesianX

Session Overview

GOAL

End-to-end coupled-RoPE K-cache polar derotate.
Match f16 quality at 3 bpw.

OUTCOME

TBQX3_1 **3.625 bpw · 4.74x compression**.
Qwen3-14B math bench **beats f16** (37.1% vs 34.3%).

MATH BREAKTHROUGH

Tangent Residual — "residual lives on tangent" insight.
4.25 bpw Direct Signs → 3.625 bpw.

HARDWARE MYSTERY

DGX Spark silent freeze root cause: **85°C thermal cutoff**.
All software hypotheses were red herrings.

TBQX3_1 Block Structure 3.625 bpw

Store in pre-RoPE (polar) domain + restore via re-rotation. Decouple RoPE coupling before quantization.

FIELD	SIZE	ROLE
`d_r`	2 B	Rayleigh σ estimate (per-block scale)
`qr[24]`	24 B	3-bit Rayleigh Lloyd-Max r indices (64 pairs)
`qphi[24]`	24 B	4-bit uniform φ indices (angular π/8 resolution)
`qtan[8]`	8 B	**1-bit tangent residual** (NEW) · φ error ±π/16

13 files modified: ggml.h, ggml-common.h, cpy-utils.cuh, fattn-common.cuh, fattn-vec.cuh, set-rows.cu, fattn.cu, ggml-cuda.cu, ggml.c, arg.cpp, common.cpp, CMakeLists, new template instances

🔑 Tangent Residual Math Breakthrough

Observation: After Rayleigh Lloyd-Max, residual error lies almost entirely along the tangent direction `(-sin φ̂, cos φ̂)`

ANALYTIC MAGNITUDE

`r · π/16` — half-width of a 3-bit cell.
No per-block metadata required.

DIRECTION IS FREE

Obtained by coordinate swap of `cos/sin` already computed during dequant.

COST

1 bit per pair (qtan[8]) + 2 FMA ops on read.

EFFECT

φ error ±π/8 → **±π/16** (22.5° → 11.25°).

3 Discarded Variants removed

VARIANT	BPW	REASON FOR REMOVAL
TBQXP3_1	4.25	Direct Signs residual. **Tangent Residual achieves same quality with 1 bit**. Overpaid — removed.
TBQX2_1	2.625	2-bit quadrant angle. π/2 resolution below attention top-1 margin. Qwen3-14B output collapsed to "Hangul" repetition. Removed.
WHT + Polar	—	Per-cell Q WHT costs **10× more**. Tangent Residual fixes the same drift cheaper. Abandoned.

Polar vs WHT cannot combine: polar preserves angular structure (math reasoning), WHT disperses outliers (prose). Different optimization domains.

Math Bench Results

Qwen3-14B · seed 1234

CONFIG	KV BUFFER	T/S	MATH /35	COMPRESSION
f16 / f16	6400 MiB	24–25	12 (34.3%)	1.00x
tbqx3 / tbq3	**1350 MiB**	21–22	13 (37.1%)	**4.74x**
tbq3 / tbq3	1250 MiB	21–22	10 (28.6%)	5.12x

Multi-seed average: tbqx3 **vs f16 ≈ +20%**, vs tbq3 ≈ +35%. seed 52 was a tie.

Cyrillic drift cured: Pauli/Schrödinger transliteration queries no longer mix Cyrillic — pure Korean output.

Pauli Transliteration Test qualitative

- **f16 / f16:** "바울리" — model limit (Korean only)
- **tbqx3 / tbq3 (bare):** "와оль프강 폴리" — **Cyrillic drift**
- **tbqx3 / tbq3 + tangent:** "와인하르트 폴리" — drift gone, still model limit

Conclusion: Pauli test is ultimately bounded by model capability, so it's unsuitable as a quantization quality metric. Subsequent focus shifted to math bench.

DGX Spark Silent Freeze Resolved

hardware

Previous hypotheses **all red herrings**: build_rope_shift / set_rows local memory / memdesc exhaustion

ACTUAL CAUSE

GB10 SoC 85°C firmware cutoff.

A silent freeze invisible to software.

FIX

Hardware voltage reduction → steady-state **low 70°C**.

Lesson: Silent hardware failure is a first-class failure mode. **Check temperature before software bisection**.

Paper Update EN + KO

- **Rayleigh Lloyd-Max subsection** new — analytic derivation + σ estimation
- **Tangent Residual subsection** new — math derivation + figure
- **Evaluation:** Phase 1-4 plan → measured-result tables × 3
- **Research Roadmap: Dead Ends section** — TBQXP3_1 / TBQX2_1 / WHT+polar / calibration / thermal
- **Typography:** pdflatex → `xelatex + fontspec(TeX Gyre Termes) + microtype + xeCJK/kotex`
- **Korean rewrite** — removed English-direct-translation feel, natural Korean prose
- PDF **12 pages (EN), 10 pages (KO)**, linked from 4 READMEs

Issue #19 Response

existential challenge

cutlerbenjamin1-cmd attack: "All TBQ variants lose to q4_0 on self-cosine / Spearman / top-5, and are 30–70% slower"

RESPONSE STANCE

- Acknowledge q4_0 competition head-on
- Admit evaluation coverage gap
- Request harness sharing
- Request v1.6.0 TBQX3_1 re-measurement

TECHNICAL CORRECTION

- QJL diagnosis fixed (tbq_signs hardcoded, tbq4_1 no QJL)
- v1.5.2 half2→float fix retained
- head_dim 128 only verified, 256/512 unvalidated

Release CI Workflow Fix resolved

Symptom: libggml-cuda.so undefined reference to `cuMemMap`, `cuMemCreate`, `cuDeviceGet` ...
Root cause: libcuda stub lives outside standard path in nvidia/cuda devel image (`/usr/local/cuda/lib64/stubs/`)

- **1st attempt (failed):** `CMAKE_EXE_LINKER_FLAGS="-L/stubs -Wl,-rpath-link,/stubs"` — rpath-link is meaningless without DT_NEEDED
- **2nd attempt (worked):** Docker script adds `ln -sf /stubs/libcuda.so /usr/lib/libcuda.so.1` — CMake find_library + ld auto-discover

Result: Run #24422871815 — **all Windows 6 + Linux 5 + aarch64 successful**, v1.6.0 release uploaded **12 binaries**.

Commit Summary

9 commits

HASH	DESCRIPTION
`c7ba61d46`	feat(tbqx3): tangent residual correction + remove failed variants
`1e84ec35a`	docs(paper): Polar Derotate paper EN/KO + README links
`2e3c26c73`	docs(paper): fix PDF typography (fontspec + microtype + table*)
`46b93143a`	docs(paper): fix column overflow (Rayleigh, verbatim, tangent eq)
`2fb6c5cd6`	chore(paper): remove tracked intermediates, add .gitignore
`b7aaf0086`	docs(readme): note TBQX3_1 is VEC-only, MMA port pending
`9c54860fe`	docs(paper-ko): soften stiff Korean prose
`97a199fd0`	ci(linux): symlink libcuda stub (final fix)

Key Lessons retrospective

- **Math breakthroughs are simple** — Tangent Residual was a one-sentence insight ("residual lives on the tangent"). 4.25 bpw → 3.625 bpw from one line of math
- **Failed research has documentation value** — 5 dead ends + thermal episode narrowed the design space
- **Silent hardware failure is first-class** — check temperature before software bisection
- **Polar vs tbq are different optimization domains** — polar = angular (math), tbq-WHT = outlier (prose). Cannot combine (per-cell WHT is 10× cost)
- **Paper prose needs to be rewritten, not translated** — direct English-to-Korean translation fails; write from scratch in natural Korean

Next Work

TBQX3_1 MMA PORT

Currently vec only 0.87× f16. MMA port restores full speed. **Top priority**.

Q4_0 DIRECT BENCH

Issue #19 challenge. Validate project's raison d'être.

HEAD_DIM DETERMINISM

256/512 unvalidated. Audit V_DOT2 undef + set_rows σ reduction.

CI DETERMINISM REGRESSION

Add automated regression test (Issue #19 suggestion).

PPL ERROR BAR RE-MEASURE

Run again with proper error bars.

UPSTREAM REBASE

24 commits behind. FA-side changes may conflict.